TEXORA · LECTURE SHEET

Fibre Tensile Strength

A yarn can never be stronger than the fibres inside it allow. This sheet covers single-fibre and bundle testing, the Pressley and Stelometer methods, and the units — tenacity above all — that make strength figures comparable between fibres of different fineness.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 8 · LEVEL 3 TERM 1

1. Quantities and units

- **Breaking load** — the force at which the specimen breaks (cN, gf, N). On its own it favours coarse fibres; it is not a quality measure until fineness enters.
- **Tenacity (specific strength)** — breaking load divided by linear density. Units: cN/tex, or the older g/tex (numerically: 1 g/tex ≈ 0.981 cN/tex — treat them as near-equal but never mix them in one table without labelling).

$$\text{Tenacity (cN/tex)} = \text{breaking load (cN)} / \text{linear density (tex)}$$

- **Breaking length** — the length of fibre that would break under its own weight; numerically equal to tenacity in g/tex expressed in km (Rkm). A legacy cotton/wool unit still met in trade.
- **Elongation at break %** — extension at break as a percentage of the original (gauge) length. Cotton fibres break at about 6–9 %; wool at 25–35 % (wet more); polyester at 15–30 % by type.
- **Stress–strain behaviour** — beyond the break point, the curve's shape (initial modulus, yield, work of rupture) governs drafting behaviour and fabric performance; bundle tests average it, single-fibre tests reveal it.

Strength depends on test conditions: **gauge length** (longer specimens are weaker — more weak places per specimen: the "weakest link" effect), **rate of loading** (constant-rate-of-extension, CRE, is the modern standard), and **moisture** (cotton gains about 10–20 % strength from dry to wet; viscose loses roughly half). All three belong in every report.

2. Single-fibre testing

Individual fibres are mounted between jaws (often on a paper card window that is cut away after clamping) and extended to break on a sensitive CRE tester. The method gives true tenacity and elongation distributions — mean, CV, weakest fibres — and is the route for research and for fibres (wool, synthetics) where bundle behaviour hides the truth. Its costs: hundreds of fibres for a stable

mean (fibre strength CV is typically 25–40 %), skilled mounting, and care that jaw breaks (specimens breaking at the jaw face) are rejected, since they measure the clamp's damage, not the fibre.

WORKED EXAMPLE — SINGLE-FIBRE TENACITY

A cotton fibre of linear density 1.7 dtex (= 0.17 tex) breaks at 6.0 cN.
Tenacity = 6.0 / 0.17 = **35.3 cN/tex** (≈ 36 g/tex) — a strong cotton fibre; typical cotton fibre tenacities run about 20–45 cN/tex by variety and maturity. Note how the same 6.0 cN on a coarse 0.30 tex fibre would give only 20 cN/tex: breaking load alone would have called the two fibres equal.

3. Bundle testing — Pressley and Stelometer

For cotton, the commercial tradition tests a **parallel bundle**: many fibres clamped together, broken as one. Bundle results predict spinning performance better than single-fibre means (a yarn is itself a bundle) and are far faster.

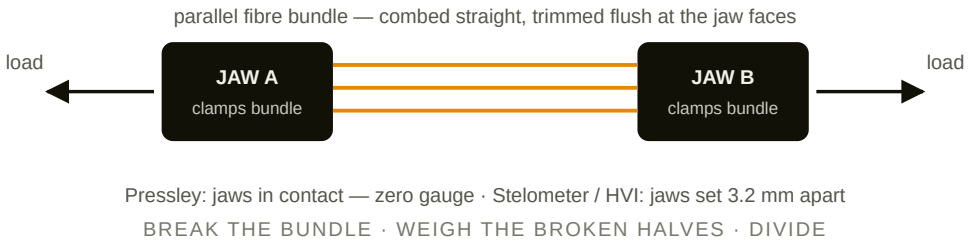


Figure 1 — Bundle tensile testing. The bundle is broken as one specimen; its mass (from the weighed halves) converts the breaking load into an index or a tenacity.

3.1 Pressley tester

A small parallel tuft is combed straight, clamped between jaws at zero gauge (the jaws touch, so all fibres are gripped over the same section), the overhanging ends are trimmed level with a razor, the bundle is broken by a rolling-weight pendulum, and the broken bundle halves are weighed. Hence:

$$\text{Pressley index (PI)} = \text{breaking load (lb)} / \text{bundle mass (mg)}$$

WORKED EXAMPLE — PRESSLEY INDEX

A bundle breaks at 18.0 lb and the bundle halves together weigh 4.8 mg.
PI = 18.0 / 4.8 = **3.75**. As a guide, Pressley indices below about 3.2 indicate weak cotton, roughly 3.6–4.0 is a sound commercial range, and values above about 4.2 are strong. The index is an instrument-specific ratio (load per unit bundle mass at zero gauge); it ranks cottons reliably but is

> converted to bundle tenacity only through the calibration for the instrument concerned — quote the index itself, with the method, rather than a converted figure of uncertain origin.

3.2 Stelometer

The Stelometer also breaks a combed bundle, but at a **3.2 mm (1/8 in) gauge** with a pendulum loading at a controlled rate, and reports bundle tenacity directly in g/tex together with bundle elongation %. The small gauge admits the weak-place effect a little, bringing bundle results closer to spinning reality; HVI (Topic 9) uses the same 3.2 mm gauge bundle principle at machine speed.

FEATURE	SINGLE FIBRE	PRESSLEY BUNDLE	STELOMETER / HVI BUNDLE
Specimen	One fibre	Parallel bundle, 0 gauge	Parallel bundle, 3.2 mm gauge
Result	Tenacity + elongation, full distribution	Pressley index	Tenacity (g/tex) + elongation %
Tests for a mean	Hundreds	~10 bundles	~10 bundles / automatic
Best use	Research, wool & synthetics	Classical cotton strength grading	Routine cotton specification

4. What fibre strength does — and does not — predict

- Yarn strength rises with fibre tenacity, but yarn tenacity is always well below fibre tenacity (commonly 40–70 % translation in cotton yarns): fibres lie at angles, slip as well as break, and the yarn fails at its thinnest place.
- Bundle strength interacts with fineness and length: at equal bundle tenacity, the finer, longer cotton usually spins the stronger yarn because its cross-section holds more, better-overlapped fibres.
- Immature fibres (Topic 5) are weak fibres — a low bundle result on a fine cotton is often a maturity problem in disguise.
- Damage shows directly: over-scorched, chemically tendered or weather-damaged cotton loses bundle strength before it loses length or colour — strength is the sensitive damage detector (used again for tendering tests on processed goods).

5. Precautions

- Condition specimens (65 % RH, 20 °C); test cotton strength unconditioned and the day's humidity writes the report.
- Comb bundles truly parallel and clamp the full width; skewed bundles break progressively and read low.
- Trim zero-gauge bundles flush; long overhangs change the effective specimen.
- Reject jaw breaks in single-fibre work; report gauge length, loading rate and number of tests with every figure.

Glossary

Tenacity — breaking load per unit linear density (cN/tex or g/tex). **Breaking length (Rkm)** — length at which a fibre would break under its own weight; numerically ≈ tenacity in g/tex, in km. **Gauge length** — the free specimen length between jaws at the start of the test. **Pressley index** — bundle breaking load (lb) ÷ bundle mass (mg) at zero gauge. **CRE** — constant rate of extension testing. **Jaw break** — a break at the clamp face, rejected as invalid. **Work of rupture** — energy absorbed to break, the area under the load–extension curve.

Key points

• Compare fibres by tenacity, never by breaking load alone.
• Gauge length, rate and moisture all move the result — report them or the number is unusable.
• Cotton routine = bundle tests: Pressley (0 gauge, index) and Stelometer/HVI (3.2 mm gauge, g/tex + elongation).
• Fibre strength sets the ceiling for yarn strength; translation into yarn is partial and depends on length and fineness too.
• Strength loss is the earliest measurable sign of fibre damage.

Practice

1. A fibre of 2.2 dtex breaks at 7.7 cN. Find its tenacity in cN/tex.
2. A Pressley bundle breaks at 16.8 lb and weighs 4.2 mg. Find the Pressley index and classify the cotton.
3. Why does the same cotton test weaker at 10 mm gauge than at zero gauge in single-fibre testing?
4. Cotton and viscose yarns are to be strength-tested wet. Predict the direction of change for each fibre and give the structural reason for the viscose result.

Answer key

1. 2.2 dtex = 0.22 tex. Tenacity = 7.7 / 0.22 = **35.0 cN/tex**.
2. PI = 16.8 / 4.2 = **4.00** — at the top of the sound commercial range; a strong cotton.
3. Weakest-link statistics: a longer specimen contains more weak places, and the test finds the weakest of them. The mean breaking load therefore falls as gauge length rises (and elongation at break spreads wider).
4. Cotton becomes **stronger** wet (better load sharing as swollen fibres pack and hydrogen bonding redistributes). Viscose becomes markedly **weaker** wet — its low degree of polymerisation and large amorphous fraction let water break the inter-chain bonds that carry load, so the wet fibre retains only about half its dry strength.
